Chapter 2 Exam Questions

1. (Multi-Answer Question) Which of the following statements about Huawei FusionSphere virtualization suite are **false**?
   A. Huawei FusionSphere virtualization suite is an industry-leading virtualization solution, which significantly improves the data processing efficiency of data center infrastructure.
   B. Huawei FusionSphere is open-source compatible.
   C. Huawei FusionSphere is used only for the virtualization of compute, storage, and network resources on servers.
   D. Huawei FusionSphere can interconnect with third-party management platforms.

2. Which of the following can implement computing virtualization and improve resource utilization in the era of cloud computing 1.0?
   A. KVM
   B. OpenStack
   C. Hyper-V
   D. Xen

3. （T or F） The virtualization technology can virtualize a physical server into multiple VMs, improving the hardware performance of the physical server.

4. （Multiple） Which of the following statements about Virt-manager are true?
   A. Virt-manager is a graphical management tool for KVM VMs.
   B. Virt-manager is a KVM VM management command set.
   C. Virt-manager is an open-source KVM VM management tool.
   D. Virt-manager is a commercial KVM VM management tool.

5. (Multiple） Which of the following are true about KVM virtualization?
   A. The host OS can be Linux.
   B. The host OS can be Windows.
   C. The guest OS can be Linux.
   D. The guest OS can be Windows.

6. (Single） The KVM virtualization architecture includes the KVM module, QEMU, and Libvirt. Which of the following statements about these modules is false?
   A. The KVM module is responsible for creating vCPUs and allocating virtual memory.
   B. The QEMU module is responsible for simulating I/O operations.
   C. Libvirt and QEMU can form a virtualization solution.
   D. KVM and QEMU can form a virtualization solution.

7. （**T** or F） Virtualization features include partitioning, isolation, encapsulation, and independence.

8. (T or **F**) Virtualization includes computing virtualization, storage virtualization, and network virtualization, which correspond to elastic cloud servers (ECSs), Elastic Volume Service (EVS) disks, and cloud networks in cloud computing, respectively.

9. Which is an open source virtualization technology?
   - **A. KVM**
   - B. Fusion compute
   - C. Hyper-V
   - D. V-sphere

10. (**T** or F) A VM has a configuration file that record the specification and the name of VM.

11. Which is computing virtualization?
   - **A. I/O virtualization**
   - **B. CPU virtualization**
   - **C. Memory virtualization**
   - D. Network virtualization

12. The main reason of rapid development in virtualization is because VM outperform the physical machine and has wider applications. (**False)**

13. The main goal of developing virtualization is.
   - **A. Make full use and utilization of hardware resources.**

14. The part of FusionCompute play a role as **a host machine** is.
   - **a. CNA**
   - b. VRM
   - c. KVM
   - d. Libvirt

15. The hypervisor used to monitor the status of host OS. (**False)**

16. In virtualization, only the hypervisor (VMM) can access the hardware resources. (**False)**

17. Which are used to allocate memory and CPU to VM in virtual environment?
   - **A. Hypervisor**
   - B. Guest OS
   - C. Host OS
   - **D. VMM**

18. If one VM infected by virus or get fault, the other VM's will not affected, which virtualization feature reflect this situation?
   - **A. Isolation**
   - B. Encapsulation
   - C. Independence
   - D. Partition

19. VM consist two files: configuration and data file, this allow migrating or moving the VM to another host for running on it, the features reflect this are.
    A. Isolation
    B. Encapsulation
    C. Independence
    D. Partition

20. The best advantage of virtualization is?
    a. VM migration between hosts
    b. Multiple VMs can run concurrently on a physical host
    c. Multiple VMs can run concurrently on a host OS
    d. In virtualization, The cpu utilization will be 60%
21. Cloud computing and virtualization are inseparable from each other. **False**

22. Open source cloud tools need an open source virtualization technology. **False**

23. Which is true about KVM?

    a. KVM is an open source
    b. KVM is type 2
    c. KVM is module in Linux kernel
    d. KVM is type1

24. Which are true about Xen and KVM?
    a. KVM and Xen are open source virtualization solution
    b. Domain 0 and Domain U are concepts in KVM
    c. KVM and Xen can run in the same host
    d. Xen VM can be converted to KVM VM

25. Which of the following are a KVM management tools?
    A. Libvirt
    B. Virsh command set
    C. Virt-manager
    D. QEMU

26. Which are operations in FusionCompute?
    A. Create VM
    B. Delete VM
    C. Clone VM
    D. Convert external VM to FusionCompute VM

27. Tools that used to create VMs in FusionCompute are
    A. Using GUI
    B. Running commands

28. Which are true about Libvirt management tool?
    A. Is an open source management tool
    B. Is an API developed using C language

C. Can be called to operate VMs of another languages
D. Is a part of KVM architecture, but it is not indispensable part
E. Can be used by other virtualization solutions

29. In FusionCompute the default alarm levels do not include()
A. Critical
B. Moderate
C. Major
D. Minor
E. Warning

30. The virtualization can virtualize a physical server into multiple VMS, improving the **hardware performance** of the physical server. **False**

31. In FusionCompute tools installed for VMs are irrelevant to the OS types and versions of the VNs. **False**

32. Which is **true** about virtualization?
A. Applications that can run on physical machines can also run on VMs.
B. VMs with different virtualization technologies can be manually converted.
C. Virtualization improves the unification of hardware resources.
D. Virtualization and non-virtualization resources cannot coexist in one datacenter

33. Which are **false** about virtualization?
A. Virtualization at IaaS layer consists of compute, storage and network virtualization
B. Computing virtualization includes CPU, Memory and I/O virtualization
C. Storage virtualization refers to distributed storage
D. Network virtualization refers to distributed virtual switches

34. Computing virtualization can be classified into type 1 and type2 or full virtualization and para virtualization from different perspective. **True**

35. Which belongs to computing virtualization?
A. CPU virtualization
B. Memory virtualization
C. Disk virtualization
D. Network virtualization
E. I/O virtualization

36. Which of the following about hypervisor is true?
A. Hypervisor is virtualization
B. A hypervisor is an OS
C. A hypervisor is a key technology in computing virtualization
D. Hypervisor contains a target number of drivers. Therefore, drivers do not need to be installed for hypervisor.

37. If KVM uses Virtio which are true about the I/O forwarding process?
A. The I/O requests need to be notified by the KVM module
B. The I/O requests are directly sent by the frontend driver to the backend driver.
C. The forwarding efficiency of I/O requests increases.
D. The I/O requests are not forwarded through QEMU.

38. With Virtio, the NIC transmission rate of Windows VM is 10 Gbit/s. **True**

39. Which of the following are true about Virtio?
A. Virtio is an I/O paravirtualization solution.
B. Virtio is an I/O full virtualization solution.
C. Virtio improves I/O forwarding efficiency
D. After Virtio is used, QEMU is not required.

40. Which of the following about VM login are true?
A. If the VM runs the Windows os, the administrator can log in to the VM using a remote desktop.
B. If the VM runs the Linux os, the administrator can log in to the VM using a remote desktop.
C. The administrator can use VNC to log in to the VM that runs any type of OS.
D. If a VM doesn't have an IP address, the administrator can log in to the VM using VNC. VNC login is not dependent on the network.

41. Which of the following are true about the encapsulation feature of virtualization?
A. Multiple VMs can concurrently run on a physical host.
B. If a VM has multiple disks, each disk generates a corresponding disk file.
C. The VM configuration file contains information such as the CPU type and quantity
D. VMs can be migrated from one physical host to another

42. Virtualization applicable to which of the following scenarios?
A. A company has 10 service systems. The resource usage rate of the host running the service systems is less than 20%
B. All employees in a company almost use the same office desktop applications.
C. A scientific research institute uses a customized operating system to ensure the confidentially of scientific research results
D. An enterprise requires a high load and compute intensive environment

43. Which of the following are true about the relationship between cloud computing and virtualization?
A. Virtualization is one of the technologies to implement cloud computing.
B. Virtualization is a technology. Cloud computing is a service mode. Virtualization promotes the development of cloud computing
C. Without cloud computing, virtualization has no value.
D. Without virtualization, cloud computing has no value.

44. The reasons why computing virtualization can improve application reliability are?

A. VM that run applications are decoupled from hardware, enabling VMs to migrate between physical hosts.

B. The computing virtualization enables multiple applications to concurrently run on the same physical host.

C. The computing virtualization can automatically adjust VM specification.

D. The computing virtualization can automatically allocate VMs that run applications to physical hosts with sufficient resources.

45. The CPU virtualization technologies classified into: full visualization, Para virtualization, software visualization and hardware assisted visualization. **False**

46. A user applies for an EVS disk in HUAWEI CLOUD. This service mode is()

A. IaaS

B. PaaS

C. SaaS

D. DaaS

47. Which of the following are features of cloud computing 1.0 era?

A. Focus on virtualization products

B. Standardized and automates resource services

C. Support agile application development and lifecycle management

D. Adopts virtualization technologies to improve resource utilization

48. In FusionCompute , a VM running Linux is created in this scenario, the **host OS runs in**( )

A. VRM

B. Linux VM

C. CNA

D. VNA

49. Which of the following are true about the type 1(bare metal) computing virtualization?

A. Applications can directly run on VMM.

B. Applications and VMs can directly run on the host OS.

C. The hypervisor directly runs on physical hardware

D. Applications cannot directly run on the host machine.

50. Which of the following is true about the type2 (hosted) computing virtualization?

A. VMM runs on the host OS

B. VMM directly runs on the host machine

C. VMM runs on the guest OS

D. VMM runs on the guest machine

51. Which of the following are true about the relation between host and guest machines?

A. One host machine can run multiple guest machines.

B. One guest machine can use resources of multiple host machines
C. One host machine can run only one hypervisor
D. One guest machine can only run on the host machine where it is created.

52. Open source virtualization technologies are all type 1 virtualization. Closed source virtualization technologies are all type 2 virtualization. **False**

53. In computing virtualization, the OS running in a VM is referred to as the host OS. **False**

54. The open source cloud computing management platform needs to be implemented by open source virtualization technologies. **False**

55. QEMU function as hardware emulator in KVM. It is a unique technology of the Linux OS. **False**

56. KVM is an open source full virtualization solution. **True**

57. In Huawei FusionCompute, users can manage VMs in which of the following modes?
A. Log in to the CNA and run commands for management
B. Log into the VM for management
C. Log in to the VRM and use the GUI for management
D. Log in to the CNA and use the GUI for management

58. Which of the following about the Huawei FusionCompute architecture is **false**?
A. CNAs can manage the network, storage and computing resources of a cluster.
B. VRM provides a management portal for administrators.
C. CNAs provide computing resources for VMs.
D. VRM can be deployed on a CNA or a dedicated server.